

LLM-Based Paper Marking and Automated Grading

Research background for exam-automark evaluation design

Scope. This survey covers automated grading of constructed responses: essays, short answers, programming submissions, mathematical work, and handwritten exam responses. Multiple-choice benchmark scoring is out of scope except where it informs evaluation methodology.

sci-brain run. Source set built on 2026-07-19 with the installed sci-brain `/survey` and `/download-ref` workflow. The active KB is `.knowledge/`: 18 grading-focused references in this report, 32 total BibTeX entries across both surveys, 32 downloaded arXiv PDFs, and 32 rendered local markdown files. No grading model calls were run.

1. Abstract

Automated grading has evolved from feature-based essay scoring to transformer models, short-answer grading benchmarks, LLM-as-a-judge protocols, and vision-capable LLMs for handwritten mathematical work [1], [2], [3], [4]. The strongest recent theme is not full replacement of human graders; it is carefully constrained human-in-the-loop automation [5], [6]. LLMs can reduce workload, draft feedback, expose rubric ambiguity, and support consistency checks, but the literature repeatedly raises concerns about calibration, bias, grading difficulty, prompt sensitivity, OCR or transcription failure, and high-stakes validity [7], [8], [9]. For exam-automark, the most useful pattern is a reproducible, rubric-item-level framework with anonymized data boundaries, strict output schemas, severe-error metrics, and teacher review gates.

2. Background

Automated essay scoring (AES) is often traced to Ellis B. Page’s Project Essay Grade work in the 1960s, but this historical citation should be rechecked before use in a formal paper. Modern AES has moved through feature engineering, statistical models, neural architectures, transformers, and feedback-oriented systems [10]. A parallel line, automatic short-answer grading (ASAG), focuses on short constructed responses where semantic equivalence, partial credit, and rubric granularity matter more than global writing quality [1], [11].

LLMs change the design space because the same model can read a rubric, compare a student answer with reference material, produce a score, and explain a decision. That flexibility is also the main risk. LLM grading is prompt-sensitive, non-deterministic unless tightly configured, prone to surface fluency effects, and often hard to calibrate against human raters. Several recent studies find promising performance in constrained formative settings, while others report that direct replacement of human grading is not yet reliable.

exam-automark sits in the practical middle. It does not need a grand claim that LLMs can grade all papers. It needs a reproducible framework for testing whether a specific grading skill, rubric, model, and data preparation pipeline can match teacher marks on a bounded assessment while flagging cases that require human review.

3. Taxonomy Of Method Categories

3.1. Classical Automated Essay Scoring

Classical AES systems score longer essays using features such as length, grammar, syntax, vocabulary, organization, and prompt relevance. Recent surveys on AES feedback and argumentative writing show a shift from holistic scores toward diagnostic feedback and trait-level evaluation [10], [12]. The lesson for exam-automark is caution: surface-quality features can correlate with marks but may miss conceptual correctness, originality, and domain-specific reasoning.

3.2. Automatic Short-Answer Grading

ASAG systems grade brief natural-language answers. Older approaches use word overlap, semantic similarity, clustering, textual entailment, and supervised classifiers [11]. Deep-learning surveys organize recent work around embeddings, sequential models, and attention-based transformer methods [1]. Several papers emphasize that hand-engineered features still complement neural representations, especially when datasets are small. Feedback-oriented ASAG work also treats the explanation of a mark as part of the grading output rather than a separate afterthought [13].

This is highly relevant to exam-automark because many exam subquestions are short constructed responses with partial-credit rubrics. It suggests that a grading skill should not rely only on an LLM’s generic judgment; it should also use rubric clauses, reference solutions, item-level labels, and deterministic post-checks.

3.3. LLM Rubric Graders

LLM grading studies ask models such as GPT-4, GPT-4o, ChatGPT, or Llama to score student answers or essays from rubrics. Some work reports near-human agreement in narrow formative contexts; other work finds weak reliability, conservative

or harsh grade distributions, and poor alignment with nuanced human grading criteria [5], [8], [14], [15]. Multi-trait specialization for AES decomposes writing quality into separate traits, which is analogous to rubric-item grading [16].

For exam-automark, direct LLM graders should be treated as candidate graders whose outputs must be validated, not as authoritative scorers.

3.4. LLM-as-a-Judge And Evaluation Bias

LLM-as-a-judge work is not education-specific, but it supplies useful warnings. G-Eval and MT-Bench show that strong LLM judges can approximate human preferences on some open-ended tasks [3], [17]. FairEval and related work document position bias, verbosity bias, and self-enhancement bias [7]. In grading, analogous biases could appear as preference for longer solutions, polished language, or answers that resemble the model’s own reasoning style.

3.5. Human-In-The-Loop And Guideline Optimization

Recent grading frameworks increasingly include human review. GradeOpt uses additional LLM agents to reflect on grading errors and optimize guidelines [18]. Human-in-the-loop handwritten-math workflows combine scanning, anonymization, multi-pass scoring, consistency checks, and mandatory human verification [6]. These are the closest cousins to exam-automark: they optimize workload and consistency without pretending that model output is final in all cases.

3.6. Multimodal And Handwritten Exam Grading

Vision-capable LLMs make handwritten exam marking possible in principle, but the literature is still early. Studies on handwritten mathematics identify transcription failures, image quality, equivalent-expression handling, and rubric misapplication as major error modes [4], [9]. This matches exam-automark’s need to separate document parsing quality from grading-skill quality.

4. Representative Work

Work	Year	Setting	Finding or role	Implication
Project Essay Grade / Page	1960s	Essay scoring	Early AES history; needs verification for exact bibliographic details	Useful historical context, not direct design evidence.
Suzen et al. [11]	2018	Short-answer grading	Text mining and clustering for short answers and feedback	Simple similarity features can remain useful baselines.
Haller et al. [1]	2022	ASAG survey	Deep-learning ASAG from embeddings to transformers	Combine learned representations with engineered features and rubric context.
Wang et al. [12]	2022	Argumentative writing survey	Trait-focused evaluation beyond holistic score	Rubric-item feedback is more useful than only total marks.
Jong et al. [10]	2023	AES feedback review	Feedback is central for AES as a learning tool	Feedback quality needs separate evaluation.
Henkel et al. [14]	2023	LLM short-answer reading comprehension	GPT-4 reached high agreement on a novel formative dataset	Promising under constrained settings with expert labels.
Schneider et al. [5]	2023	LLM short textual answers	LLMs can support validation but need human oversight	Use LLM graders as complementary evidence, not final authority.
Lee et al. [16]	2024	Zero-shot essay scoring	Multi Trait Specialization decomposes scoring into traits	Map essay traits to rubric-item scoring.
Mansour et al. [15]	2024	LLM AES	Prompt choice is model- and task-dependent; feedback may be useful	Prompt experiments need frozen metrics and held-out tests.
Aggarwal et al. [13]	2024	Short-answer feedback	Automatic grading is paired with student-facing feedback	Feedback quality needs its own review target.
GradeOpt [18]	2024	LLM short-answer grading	Multi-agent guideline optimization improves behavior alignment	Relevant to automated grading-skill improvement.
ASAG2024 [2]	2024	Combined ASAG benchmark	LLM methods reach high scores but remain below human performance	Benchmark generalization remains unresolved.

Caraeni et al. [4]	2024	Handwritten math exams	Rubrics improve GPT-4o alignment, but accuracy remains too low for deployment	Multimodal grading needs review gates.
Vanhoyweghen et al. [6]	2026	Handwritten math, human-in-loop	Workflow reduced grading time while requiring verification	Closest operational pattern for exam-automark.
Levine et al. [9]	2026	Vision-capable LLM handwritten math	Many errors attributed to transcription rather than rubric application	Separate OCR/transcription evaluation from grading evaluation.

5. Comparison Table

Approach	Best fit	Required data	Strengths	Risks
Classical AES	Long essays and writing traits	Large scored essay sets	Efficient, stable, interpretable features [10]	Can reward length or fluency over content.
Traditional ASAG	Short typed answers	Reference answers and scored examples	Strong baselines; transparent similarity features [11]	Question-specific; struggles with rare valid paraphrases.
Transformer ASAG	Short answer datasets	Labeled responses per task	Better semantic representations [1]	Small-data overfitting and domain transfer issues.
Direct LLM rubric grader	Low-stakes formative or pilot grading	Rubric, reference solution, examples	Fast to prototype; can produce feedback [14], [15]	Prompt sensitivity, bias, hallucination, calibration drift.
LLM-as-a-judge protocol	Comparing outputs or feedback variants	Human preferences or rubrics	Scalable evaluation aid [3], [17]	Position, verbosity, and self-preference biases [7].
Guideline optimizer	Improving grading instructions	Error traces and human labels	Can discover rubric ambiguity and revise criteria [18]	Needs strict audit trail to avoid self-justifying rules.
Multimodal LLM grader	Scanned handwritten exams	Images, transcriptions, rubric labels	Can unify reading and marking [4], [9]	Image quality, OCR, formula equivalence, and review burden.

6. Relevance To exam-automark

Design direction. exam-automark should evaluate grading as a pipeline: document preparation, evidence extraction, rubric-item scoring, schema validation, feedback, and audit. Each stage needs its own failure categories.

Deployment stance. The literature supports LLM-assisted grading with human verification more strongly than fully autonomous high-stakes grading. Severe-error flags and manual review are design requirements, not optional polish.

Concrete lessons for the repository:

- Store only anonymized, reproducible experiment metadata in Git; keep raw submissions outside the public repo.
- Use item-level metrics, not just total-score correlation. Exact agreement, adjacent agreement, total MAE, severe-error rate, bias, and per-question deltas should all be reported.
- Separate parsing/OCR errors from rubric errors. A vision model can fail before the grading skill has a fair chance.
- Require strict output schemas and validation before using any model output in metrics.
- Compare against human-human agreement when possible. A model-human mismatch is more meaningful when human rater reliability is known.
- Use teacher review for ambiguous or high-impact cases, especially where a partial-credit decision changes the pass/fail or letter-grade boundary.
- Evaluate feedback separately from marks. A correct score with misleading feedback is still unsafe for learners.

7. Open Problems

- **Validity.** Agreement with historical marks does not prove the system measures the intended construct, especially for open-ended reasoning.
- **Fairness.** LLMs may favor fluent writing, standard dialects, longer answers, or model-like reasoning even when a rubric should reward substance.

- **Human label noise.** Single-rater grades can be inconsistent. Automated systems should not be trained to reproduce unexamined noise.
- **Ambiguous partial credit.** Rubrics often leave edge cases unresolved. These cases should become teacher-review flags and rubric revisions.
- **Benchmark generalization.** ASAG2024-style combined benchmarks help, but course-specific rubrics and handwriting still challenge transfer.
- **Multimodal robustness.** Image capture, cropping, handwriting, formulas, tables, diagrams, and units all create non-text failure modes.
- **Feedback safety.** LLMs can generate confident explanations for wrong scores. Feedback needs factual checks against the submitted work.
- **Operational audit.** Every grade should be traceable to model version, prompt packet, rubric version, input hash, validation result, and reviewer decision.

8. References

Generated from the sci-brain KB BibTeX file copied from `.knowledge/references.bib`. Historical Page/PEG context remains needs verification because it is not part of the verified arXiv/Semantic Scholar KB.

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